ECON 0100 | Exercise C2 | Taxes & Subsidies

In recent years Hogsmeade has seen a proliferation of candy shops selling smoked toffees. The Demand (marginal benefit) curve and Supply (marginal cost) curve for smoked toffee can be represented by the following relationships.

$$D : P_B = 20 - \frac{1}{2}Q_D \qquad\qquad S : P_S = 2 + Q_S$$

Due to the metabolic issues among the younger generation of wizards in the community resulting from the consumption of these high glycemic index treats, the Ministry of Magic imposed a 3 Coin tax on the sale of smoked toffee.

Q1 | One Toffee Exchange

	a) A Single Exchange Buyer's MB: 16 Coins. Seller's MC: 12 Coins.	**b) A Single Exchange** Buyer's MB: 14 Coins. Seller's MC: 12 Coins.
With no tax, will they exchange?	_____	_____
With the tax, will they exchange?	_____	_____
Total Surplus to buyer and seller:	_____	_____
Tax paid to the Ministry of Magic:	_____	_____
Total Surplus (with the Ministry):	_____	_____
Deadweight Loss:	_____	_____

Q2 | Tax Equilibrium

Use a graph and algebra to discuss the impact of the tax on the market. Include how the welfare areas changed after the policy. No need to calculate welfare measures except DWL.

Pre-Tax Price ($\hat{P}$):	_____	Pre-Tax Quantity ($\hat{Q}$):	_____
Buyer's Price (P_b):	_____	Seller's Price (P_s):	_____
Post-Tax Quantity (Q^{Tax}):	_____	Dead Weight Loss (DWL):	_____
Tax Incidence on Buyers:	_____	Tax Incidence on Sellers:	_____

Who bears more of the tax? *buyers* · *sellers* · *equal*

Q3 | Subsidy Equilibrium

Now suppose that instead of the tax, the Ministry of Magic pays a 3 Coin subsidy on every smoked toffee sold. Use a graph and algebra to discuss the impact of the subsidy on the market.

Buyer's Price (P_b):	_____	Seller's Price (P_s):	_____
Post-Sub Quantity (Q^{Sub}):	_____	Deadweight Loss (DWL):	_____